

Control of 3-dimensional motion in plasmonic nanomotors with optical pulling forces

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The transfer of momentum during light-matter interactions generates significant optical forces, which play a crucial role in various micro- and nanotechnology applications, particularly in the development of lab-on-a-chip systems [1]. In these systems, components such as pumps, valves, mixers, etc. have traditionally relied on gradient optical forces from optical tweezers, which significantly restricts the materials, sizes and geometries of the actuated objects. To address this challenge, several nanomotors based on scattering forces from asymmetric plasmonic nanoantennas or nanophotonic metasurfaces have been recently proposed [2,3]. These approaches have been relatively successful in controlling movement within the transversal plane, even to the three degrees of freedom in two-dimensional movement. However, controlling movement in the longitudinal direction remains an unsolved challenge [4].

In particular, while optical pushing forces are relatively straightforward as a result of photons recoiling on the material; the counterintuitive concept of optical pulling forces was only discovered and explored in the last decade [5]. To achieve such forces, the recoil of photons must generate a negative force, i.e., the object must collimate or squeeze the incident light. This imposes strict limitations on both the object's properties and the characteristics of the incident light. To address these challenges, large cone angle Bessel beams or pairs of light sources have been utilized, with significant efforts dedicated to enhancing their practicality and applicability [6].

Here we theoretically propose a realistic nanomotor design whose motion can be controlled both transversely and longitudinally. A suitable platform for optical pulling was first developed, based on waveguide coupling between an azimuthally polarized Bessel beam and a glass cylinder. Then, using this cylinder as the main body for the nanomotor, asymmetric plasmonic dimers were strategically placed within the dielectric cylinder to avoid strong interaction with the pulling beam. Therefore, a plane wave incidence produces lateral motion, driven by the lateral force from the plasmonic dimers, while incidence with a Bessel beam provides movement on the longitudinal axis via the optical pulling force on the cylinder. Our design enables the selection of motion modes: pulling, pushing, and lateral movements (left-right, forward-backward) by adjusting the polarization or switching between plane wave and Bessel beam illuminations. Furthermore, we demonstrate that the optical pulling force achieved in our system is highly robust against rotations, translations and Brownian motion, furthering the versatility and practical applicability of our device.

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